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The linear function

$y = kx + b$ — a straight line, with k its steepness and b where it crosses.

LESSON 141–142 | 2 × 40 MIN | MATEMATIKA 7, §61 | S8 9.2–9.3

LESSON CLOCK

15' | 25' | 20' | 20' | 5'

Two points are enough	15 min
What k and b do	25 min
Meeting the axes	20 min
Is the point on the line?	20 min
Homework	5 min

LEARNING OBJECTIVES

- Draw the graph of $y = kx + b$ from two points.
- Interpret k as the gradient and b as the intercept.
- Find where the line meets each axis.
- Decide whether a point lies on a given line.

TEXTBOOK

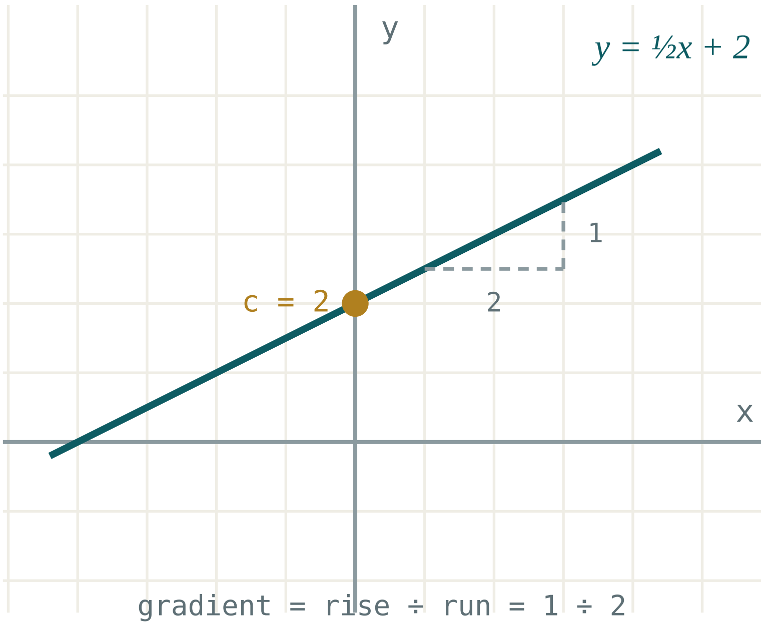
National	pp. 423–431
Cambridge	Stage 8, pp. 88–95
Workbook	Exercise 9.3

01 Explanation

Two points are enough

$$y = kx + b$$

Every such function has a straight line for its graph, and two points determine a straight line — so a table of two values is all that is needed. A third point is worth plotting as a check.



The gradient and the intercept of a line

FUNCTION	$X = 0$	$X = 1$	LINE THROUGH
$y = 2x + 1$	1	3	$(0; 1)$ and $(1; 3)$
$y = -x + 4$	4	3	$(0; 4)$ and $(1; 3)$
$y = 3x$	0	3	$(0; 0)$ and $(1; 3)$

$B = 0$ MEANS THE LINE PASSES THROUGH THE ORIGIN

Then $y = kx$ is a **direct proportion**: doubling x doubles y. Every proportional relationship met in Grade 6 was a linear function with $b = 0$.

What k and b do

COEFFICIENT	MEANING	ON THE GRAPH
$k > 0$	increasing	rises left to right
$k < 0$	decreasing	falls left to right
$k = 0$	constant	a horizontal line
b	the value at $x = 0$	where the line cuts the y-axis

The gradient is the change in y for a step of 1 in x:

$$k = \frac{y_2 - y_1}{x_2 - x_1}$$

TWO POINTS	WORKING	K
(0; 1) and (1; 3)	$\frac{3-1}{1-0}$	*2*
(0; 4) and (1; 3)	$\frac{3-4}{1-0}$	*−1*
(1; 2) and (4; 8)	$\frac{8-2}{4-1}$	*2*

THE TWO DIFFERENCES MUST BE TAKEN IN THE SAME ORDER

$\frac{y_2 - y_1}{x_1 - x_2}$ gives the gradient with the wrong sign. Fix an order — second minus first, top and bottom — and keep it.

Meeting the axes

AXIS	SET	GIVES
y	$x = 0$	$y = b$
x	$y = 0$	$x = -\frac{b}{k}$

FUNCTION	CUTS OY AT	CUTS OX AT
$y = 2x + 1$	$(0; 1)$	$(-0.5; 0)$
$y = -x + 4$	$(0; 4)$	$(4; 0)$
$y = 3x - 6$	$(0; -6)$	$(2; 0)$

THE TWO INTERCEPTS ARE THE TWO EASIEST POINTS

For a line with $b \neq 0$ they are usually the quickest pair to plot, and they lie far apart, which makes the drawn line more accurate than one built from two nearby points.

Is the point on the line?

Substitute the coordinates. If the equation becomes true, the point lies on the line; if not, it does not.

LINE	POINT	CHECK	ON IT?
$y = 2x + 1$	$(3; 7)$	$2 \cdot 3 + 1 = 7$	yes
$y = 2x + 1$	$(3; 6)$	$7 \neq 6$	no
$y = -x + 4$	$(1; 3)$	$-1 + 4 = 3$	yes
$y = 3x$	$(2; 5)$	$6 \neq 5$	no

THIS IS WHAT THE NEXT BLOCK IS BUILT ON

A pair $(x; y)$ satisfying two linear equations at once lies on both lines — that is, at their crossing point. Systems of equations, three lessons from now, are exactly that idea.

02 Worked examples

EXAMPLE 1 **Draw the graph of $y = 2x + 1$.**

1. $x = 0$ gives $y = 1$.
   The intercept.
2. $x = 1$ gives $y = 3$.
3. Plot $(0; 1)$ and $(1; 3)$ and join them.
4. Check with $x = -1$: $y = -1$ lies on the same line ✓

ANSWER

A line through $(0; 1)$ and $(1; 3)$

EXAMPLE 2 **Find where $y = 3x - 6$ meets the axes.**

1. $x = 0$: $y = -6$, so $(0; -6)$.
2. $y = 0$: $3x - 6 = 0$.
3. $x = 2$, so $(2; 0)$.

ANSWER

$(0; -6)$ and $(2; 0)$

EXAMPLE 3 **A line passes through $(1; 2)$ and $(4; 8)$. Find its gradient.**

1. $k = \dfrac{y_2 - y_1}{x_2 - x_1}$
2. $= \dfrac{8 - 2}{4 - 1}$

   Same order top and bottom.
3. $= \dfrac{6}{3} = 2$

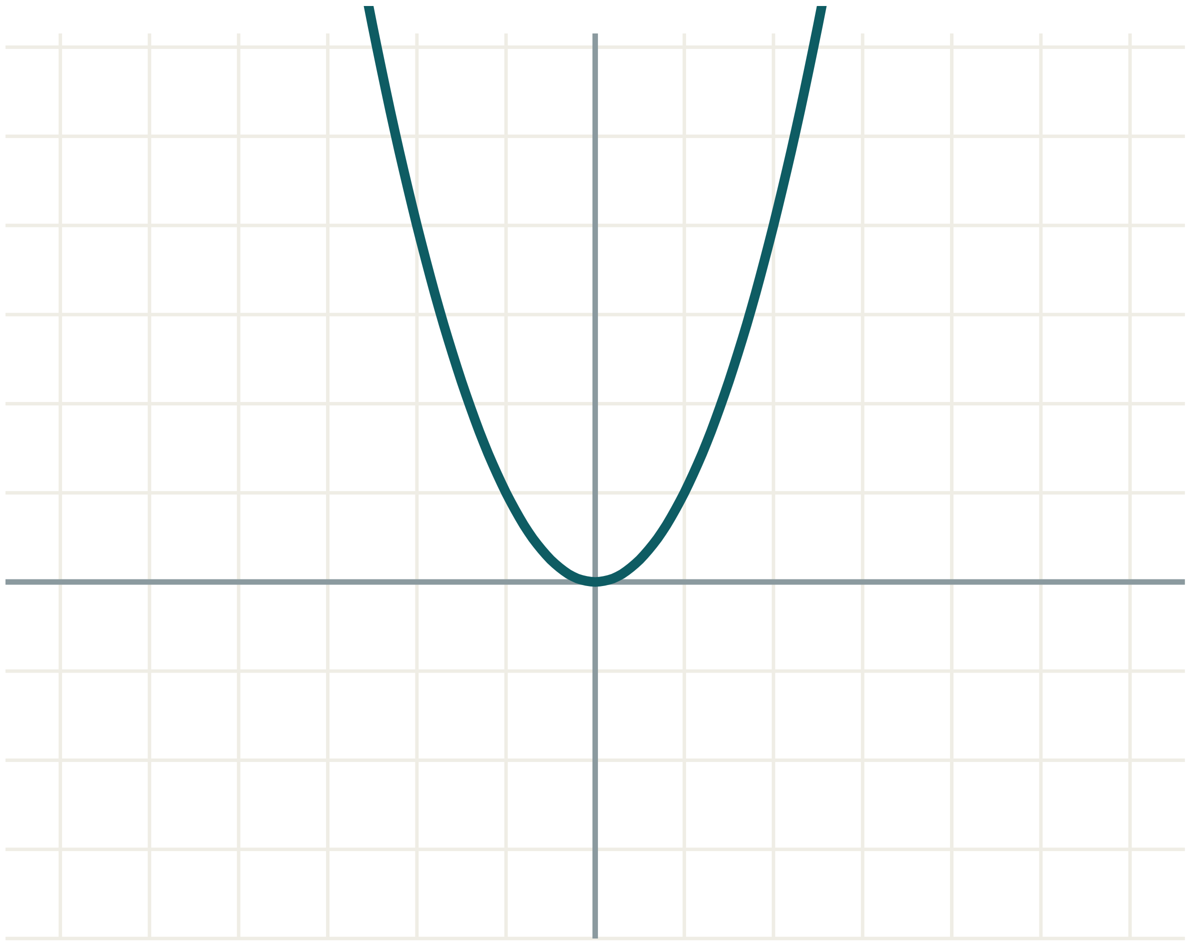
ANSWER

$k = 2$

03 Interactive model

Draw y = 2x + 1, y = 2x and y = 2x − 3 on one grid; the class sees at once that k tilts the line and b slides it, without being told.

Move *k* and *b* and watch the line



a **1**

b **0**

c **0**

equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Linear function	Chiziqli funksiya	Линейная функция
Gradient	Burchak koeffitsiyenti	Угловой коэффициент
Intercept	Kesma	Свободный член
Straight line	To‘g‘ri chiziq	Прямая
Increasing	O‘suvchi	Возрастающая
Decreasing	Kamayuvchi	Убывающая
Direct proportion	To‘g‘ri proporsionallik	Прямая пропорциональность
To lie on a line	Chiziqqa tegishli	Принадлежать прямой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The graph of $y = kx + b$ is:

- a curve
- a straight line
- a parabola
- two lines

How many points determine it?

- 1
- 2
- 3
- many

b is the value of y when:

- $x = 1$
- $x = 0$
- $y = 0$
- $k = 0$

$k < 0$ means the line:

- rises
- falls
- is horizontal
- is vertical

$y = 3x - 6$ meets Ox at:

- $(0; -6)$
- $(2; 0)$
- $(-2; 0)$
- $(6; 0)$

Is $(3; 6)$ on $y = 2x + 1$?

- yes
- no
- only if k changes
- cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $y = 2x + 1$ at $x = 0$

   ANSWER $y = 1$

2. $y = 2x + 1$ at $x = 1$

   ANSWER $y = 3$

3. $y = -x + 4$ at $x = 1$

   ANSWER $y = 3$

4. The intercept of $y = 3x - 6$

   ANSWER $(0; -6)$

5. Is $y = 3x$ increasing?

   ANSWER Yes

6. Is $y = -2x + 1$ increasing?

   ANSWER No

7. The graph of $y = 5$ is

   ANSWER a horizontal line

Medium · the standard question

7 PROBLEMS

1. $y = 3x - 6$ meets Ox at

   ANSWER $(2; 0)$

2. $y = -x + 4$ meets Ox at

   ANSWER $(4; 0)$

3. $y = 2x + 1$ meets Ox at

   ANSWER $(-0.5; 0)$

4. The gradient through $(1; 2)$ and $(4; 8)$

   ANSWER $k = 2$

5. The gradient through $(0; 4)$ and $(1; 3)$

   ANSWER $k = -1$

6. Is $(3; 7)$ on $y = 2x + 1$?

   ANSWER Yes

7. Is $(2; 5)$ on $y = 3x$?

   ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. A line through $(0; 3)$ with gradient -2

   ANSWER $y = -2x + 3$

2. A line through $(1; 5)$ and $(3; 11)$

   ANSWER $y = 3x + 2$

3. For which k does $y = kx + 1$ pass through $(2; 7)$?

   ANSWER $k = 3$

4. Where do $y = 2x + 1$ and $y = -x + 4$ cross?

   ANSWER $(1; 3)$

5. Two lines with the same k are

   ANSWER parallel

6. The graph of $y = kx$ always passes through

   ANSWER the origin

7. For which b does $y = 2x + b$ pass through $(3; 1)$?

   ANSWER $b = -5$

07 Homework

Homework — 5 tasks

Plot three points, not two — the third is your check.

1. Draw the graphs of $y = 2x - 3$ and $y = -x + 2$ on one grid.
2. Find where each of them meets the axes.
3. Find the gradient of the line through $(2; 1)$ and $(5; 10)$.
4. Decide whether $(4; 5)$ lies on $y = 2x - 3$.
5. Find b so that $y = 3x + b$ passes through $(2; 4)$.